Credit Name: CSE 2110 Procedural Programming 1

Assignment Name: GuessingGame.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

I broken down the game logic into two main tasks: repeatedly getting guesses from the user until they get it right, and passing their input into a custom helper method to evaluate whether their guess was too high or too low.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Passing arguments into a separate helper method (giveHint(userGuess, secret)) was slightly confusing at first in terms of matching parameters, but I checked our class examples to make sure the parameter data types and order matched correctly.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes, I planned to create the helper method first with the comparison conditions, and then build the main method with the while loop to repeatedly prompt the user until the secret number was matched.

How did you decide on the tools, data structures, or algorithms to use?

I used a while loop because I didn't know in advance how many tries the user would take, int variables to hold the integers, and a custom void method to keep the main method cleaner.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I wrote the giveHint() helper method first and tested basic if-else prints, then added the while loop and Scanner in main() to control the game loop.

How did you test your solution along the way to make sure it was working?

I tested with numbers higher than 4, lower than 4, and exactly 4 to verify that the hints popped up correctly and that the loop terminated properly on a win.

Overcoming Challenges

What part of the problem was the most difficult for you?

Ensuring the while condition (userGuess != secret) evaluated properly so the game didn't create an infinite loop or stop prematurely before the user won.

How did you handle moments when you felt stuck or unsure of what to do next?

I traced the loop execution line-by-line using test numbers to confirm when variables updated during each iteration.

Learning

Was there anything you learned that you think will help you with future challenges?

I got a much better understanding of how parameter passing works between main() and helper methods, as well as using condition-driven loops like while for interactive programs.